

Indemnity Payments in PES Programs: A Practical Guide

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Abstract

Introduction

This paper seeks to answer the question of, when is it optimal for a regulator to offer indemnity (loss protection) for agents who opt into a payment for ecosystem services program (PES)? In the most general sense, ecosystem services (sometimes referred to as environmental services) are the aspects of an ecosystem (or environment) that provide benefits to humanity, directly or indirectly (Millennium Ecosystem Assessment, 2005; Fisher, Turner and Morling, 2009).¹ A majority of ecosystem services (ES) fall within the realm of public goods and externalities and therefore are supplied at suboptimal levels. Worse yet, early assessments indicate that these services are degrading at rapid rates (Millennium Ecosystem Assessment, 2005).

One solution for this market failure that has been gaining popularity among scholars and practitioners is known as payments for ecosystem services (PES) (Salzman et al., 2018). PES programs facilitate the creation of contracts between ES beneficiaries (or their representative such as the government or regulator) and ES suppliers where the beneficiary agrees to pay the supplier for their provision of ES. PES contracts are voluntary programs where agents have to opt into participation which is in contrast to Pigouvian taxation. Around the world, there are now 550 PES programs with a combined annual payment of around \$36 billion USD (Salzman et al., 2018). It is true that there is no consensus on the definitions of ES (Fisher, Turner and Morling, 2009) and PES (Schomers and Matzdorf, 2013) and the disagreement for both definitions are centered on how narrow or broad each definition should

¹Direct benefits could include eco-tourism, pollination for agriculture, carbon sequestration, and general ecosystem health to prevent desertification. Examples for indirect benefits include non-use value from biodiversity, predator populations to prevent deer overpopulation, etc.

be. Throughout this paper, both ES and PES will be used to refer to the broadest definitions of both.

PES schemes are, loosely speaking, an application of the Coase Theorem and can be contracted on either ES outputs (like lower atmospheric CO₂) or ES inputs (like carbon sequestering land uses) and which one to contract on is still an active area of research (White and Hanley, 2016). PES contracts based on outputs are sometimes referred to as performance-based contracts while input-based ones are also known as action-based contracts. One central tradeoff between these two types is that contracting on outputs ensures that the buyers receives the benefits of higher ES. However, because of the variability in nature, the ES production function is inherently stochastic which means that contracting on ES outputs could put significant risk on the agent. Further, it is typically much harder to monitor ES outputs compared to ES inputs (Burton and Schwarz, 2013; Derissen and Quaas, 2013).²

Consequently, many PES programs are contracted on ES inputs rather than outputs (Jack, Kousky and Sims, 2008) with the majority of European PES programs being input-based (Wuepper and Huber, 2022). Though this approach may attenuate the risks faced by the agent, it does not necessarily eliminate it. There are many instances in which the ES inputs themselves introduce risk to the agent’s bottom line such as payments to encourage coexistence with predators. The ES input there would be an inaction, i.e., avoided revenge killing of a predator that caused damage to either property (livestock) or human life (Dickman,

²Though remote sensing is making monitoring environmental factors at large more and more accessible, it is still quite difficult to attribute individual contributions to aggregate environmental quality measures.

Macdonald and Macdonald, 2010). A more common example are conservation programs that target land use goals where risk tolerance and optimal portfolio theory plays a significant role due to the uncertainty of future profit streams (Benítez et al., 2006; De Pinto, Robertson and Obiri, 2013). Thus the central focus of this paper is on input-based PES contracts and the associated risk to agents.

Specifically, the main goals of this paper are to (i) define the necessary and sufficient conditions for when it is optimal for the regulator to offer positive indemnity payments, (ii) offer an approach to evaluate how large such gains could be, (iii) define further conditions for when it is optimal to pursue the dual objective of additionality and poverty alleviation and (iv) offer a guide to practitioners on how to estimate whether their specific context could benefit from indemnity and by how much and to estimate whether the dual objective is optimal.³ Linear pricing PES contracts defines a measurable and verifiable ES input and pays a subsidy for each unit of input. This is in contrast to optimal contracts from the mechanism design literature. Those contracts are optimal in the sense that it maximizes social welfare by explicitly accounting for the asymmetric information about agent's counterfactual ES input supply (i.e., additionality). They are a menu of pairs of ES input levels and the corresponding payment, carefully designed to create a separating equilibrium that maximizes expected welfare (Mason and Plantinga, 2013).

The PES literature does address compensation and/or insurance schemes which have pri-

³Indemnity simply means protection against financial loss/burden. Insurance is a subset of indemnity in that it specifically requires the indemnitee to pay the indemnitor for indemnity.

marily evolved from the predation context (Dickman, Macdonald and Macdonald, 2010). Though they are called either a compensation or insurance scheme, they are in fact a form of linear pricing PES when the inaction (i.e., to not pursue a revenge killing) is viewed as an ES input. To the best of my knowledge, the literature has not yet address the issue of optimal linear PES contracts when the cost of ES input provision is uncertain to both principal and agent. Nor has the literature addressed the role that “insurance” or, more accurately, indmenity against the risk of supplying ES inputs can play in achieving ES goals. The main difference between insurance and indemnity is that insurance implies that there is an insurance premium whereas indmenity is a much more general sense of protection against loss and does not require the insuree to pay a premium or cost of insurance.

The next section outlines the model detailing both agent’s and principals problems where I show that it is optimal for the regulator to offer indemnity payments as long as the indemnity elasticity of ES input supply is sufficiently higher than the pay rate elasticity of ES input supply. The next section deals with additionality and outlines conditions in which it is optimal for the regulator to pursue the dual objective of additionality and poverty alleviation.

Model

The linear PES contract is characterized by (1) an indemnity rate $I \in [0, 1]$ that is paid out for each dollar of loss ($x_i \epsilon_i$) that occurs and (2) the ES input payment rate p which is paid for each unit of ES input x_i (henceforth referred to as the pay rate). There is a cost shock ϵ_i that is distributed $(\mu_\epsilon, \sigma_\epsilon^2)$ and has support over $[0, e]$ and when multiplied with x_i produces

the loss that an agent faces for choosing x_i . I assume that only one ES input x_i is being contracted and that the input is continuous over $[0, \infty]$ so that if a cost shock occurs, the PES participant's loss equals $x_i \epsilon_i$ but receives $x_i \epsilon_i I$ in indemnity payments. If the agent does not participate in the PES program, then $x_i = 0$. Potential enrollees all have heterogeneous initial wealth ω_i and are heterogeneous in their known input cost function $g_i(x_i, \omega_i)$ and for brevity, subscripts i will be henceforth omitted when risk of confusion is low.⁴

Agents have Bernoulli utility $u(c)$ over consumption c and a budget constraint defined in (1). Thus, consumption is only stochastic when opting into the PES program by choosing $x > 0$ and deterministic if $x = 0$.⁵ For any given policy level (I, p) , there will be X units of ES input supplied in the aggregate which then engenders ES output $ES(X)$. This ES output then leads to some level of social benefit $\tilde{B}(ES(X))$ which may be random as well. From here on, denote $B = \tilde{B} \circ ES$ as the expected social benefit and is a function of the aggregate ES inputs provided, X .

Agent's Problem

Agents are expected utility maximizers who take contract (I, p) as given and chooses ES input level.

⁴Small letters will denote individual level values and big letters denoting aggregate values.

⁵There is no loss of generality since one could simply treat ω as a random variable as well.

$$\begin{aligned}
& \max_x \mathbb{E} [u(c)] \\
& \text{s.t.} \quad c = \omega + xp - g(x, \omega) - x\epsilon(1 - I) \\
& \quad x \geq 0
\end{aligned} \tag{1}$$

The ES input cost function is assumed to be increasing and convex ($g_x(x, \omega) > 0$, $g_{xx}(x, \omega) > 0$).⁶ The first order condition is given by (2) which implicitly defines the interior solution x^* .

$$\mathbb{E} \left[u'(c^*(x^*, \omega)) \frac{\partial c^*}{\partial x} \right] = 0 \tag{2}$$

Using monotone comparative statics, it is possible to sign $\frac{\partial x^*}{\partial p}$ and $\frac{\partial x^*}{\partial I}$.

Proposition 0.1. *Given contract (I, p) , the provision of ES inputs is non-decreasing in indemnity payment rate I nor is it decreasing in the payment rate p , i.e., $\frac{\partial x_i^*}{\partial I} \geq 0$ and $\frac{\partial x_i^*}{\partial p} \geq 0$.*

See proof on page 23.

Proposition 0.1 says that the risk averse agent's supply of ES inputs weakly increases with both the pay and indemnity rates. This implies that there must necessarily be a trade off faced by the regulator with a fixed budget G who maximizes expected social benefit, $B(X)$.

The natural question then arises, when is it optimal for such a regulator to offer a positive indemnity rate at the expense of offering higher p ?

⁶One could allow agents to derive positive benefits from provision of x and include a second argument into the utility. Then the RHS of (2) would instead be equal to $\mathbb{E}[u_x(c, x)]$. Further, one would have to follow more closely to Chetty (2006) in order to account for the role of complementarity between x and c in the application stage of this paper.

Planner's Problem

The social planner takes government budget G as fixed and seeks to maximize aggregate supply of ES input $X = X(x_1^*, \dots, x_n^*) = \sum_i x_i^*$. Note that the formulation of (3) excludes the welfare of the ES suppliers since the goal is not to achieve a first best outcome, rather the objective is to maximize the benefit of a public good given a fixed budget. If the regulator's budget constraint is not binding then the regulator's solution is isomorphic to the Coasean bargaining outcome and will actually achieve first best.

$$\max_{(I,p)} B(X) \quad \text{s.t.} \quad X(p + \mu_\epsilon I) \leq G \quad (3)$$

With the inclusion of indemnity payments, the government expenditure for any given PES contract (I, p) is now stochastic meaning that there are multiple ways to formulate a budget constraint for the regulator. Equation (3) formulates the budget constraint in terms of expectations where μ_ϵ is the expected value of the cost shock. Alternatively, the budget constraint can be formulated in probabilistic or extreme terms, e.g., the probability that the regulator's expenditure exceeds G is less than some threshold or the budget constraint can be formulated to ensure that the budget will not be exceeded in the event of an extreme shock (where all agents experience shock $\epsilon_i = e$).

When are indemnity payments worth it?

The formulation of (3) differs slightly from the standard linear pricing PES contracts. The standard optimal linear pricing PES contract takes $I = 0$ as given, i.e., there is never any indemnity payments and is usually so for exogenous reasons rather than a result of the regulator behaving optimally. I refer to this type of contract as the conditionally optimal linear pricing PES contract (i.e., optimal conditional on a given $I = 0$). The optimal linear pricing PES contract refers to the case where the regulator optimally chooses both I and p . The central question here is whether the regulator can achieve a higher indirect social benefit function when the artificial constraint ($I = 0$) is relaxed. In other words, when is it optimal for the regulator to share the risk of supplying ES inputs with the ES input suppliers?

Theorem 0.2. *Given the ES supply function $X = \sum_i x_i^*$, it is optimal for the regulator to offer a positive indemnity rate ($I > 0$) if and only if the marginal increase in ES inputs from an increase in I is sufficiently greater than the marginal increase in ES inputs from an increase in p . Specifically, $I^* > 0$ if and only if*

$$\frac{\partial X}{\partial I} > \frac{\partial X}{\partial p} \mu_\epsilon$$

holds when p is optimally chosen given $I = 0$.

See proof on page 24.

Corollary 0.3. *The regulator can achieve higher social value given a fixed budget by offering a positive indemnity rate if there exists $I_0 > 0$ such that at I_0 , the indemnity elasticity of (ES input) supply is sufficiently greater than the pay rate elasticity of supply. Specifically,*

$I^* > 0$ if there exists $I_0 > 0$ such that

$$\varepsilon_I > \varepsilon_p \mu_\epsilon \frac{I_0}{p}$$

holds when p is optimal conditional on I_0 where $\varepsilon_I = \frac{\partial X}{\partial I} \frac{I_0}{X}$, $\varepsilon_p = \frac{\partial X}{\partial p} \frac{p}{X}$, $\mu_\epsilon > 1$ and $\frac{I_0}{p} < 1$.

See proof on page 25.

Under both optimal and conditionally optimal linear contracts, the regulator's budget will always be exhausted. If the regulator wanted to increase the indemnity rate I from zero, then their expenditure will increase by $X\mu_\epsilon$ (assuming no supply responses) but then the pay rate p must be reduced which lowers their expenditure by X . However, increasing I and decreasing p will lead to countering effects on the supply of ES inputs X which has further impacts on the budget. Thus, Theorem 0.2 simply says that in order for this budget reshuffling to be optimal, the increase in X in response to an increase in I must outweigh the decrease X in response to the required decrease in p .

This point can be further illustrated by looking at (4) where (4a) shows that increasing I increases regulator's expected expenditure (i) through an increase in supply X and (ii) through increasing the amount paid on existing supply $X\mu_\epsilon$. Then to re-balance the budget, a decrease in p is needed leading to lower spending on existing supply equal to X and by lowering the total ES offered. Suppose that the condition from Theorem 0.2 holds with equality ($\partial X/\partial I = \partial X/\partial p \mu_\epsilon$). Then a one unit increase in I requires p to decrease by μ_ϵ for budget balance. This would result in zero change in ES input. If the condition from Theorem 0.2 holds with strict inequality, then the pay rate p would have to decrease by

more than μ_ϵ to balance the budget but will still result in a higher X . For instance, suppose $\partial X/\partial p \approx 0$ so that the required decrease in p equals $\mu + Z$ since it would have to make up for increased budget on both channels stated earlier. However, the X would clearly be higher and so this reshuffling is optimal for the regulator.

$$\frac{\partial G}{\partial I} dI = \left[\frac{\partial X}{\partial I} (p + I\mu_\epsilon) + X\mu_\epsilon \right] dI \quad (4a)$$

$$\frac{\partial G}{\partial p} dp = \left[\frac{\partial X}{\partial p} (p + I\mu_\epsilon) + X \right] dp \quad (4b)$$

Theorem 0.2 is quite important as it provides a necessary and sufficient condition for which the regulator can always extract more social value given a fixed budget G by offering indemnity payments. Essentially, whether $I > 0$ is socially optimal depends on (i) how responsive ES input supply is to changes in I relative to p and (ii) the risk of offering indemnity which is represented by the expected cost shock μ_ϵ . High values for μ_ϵ even when $\varepsilon_I > \varepsilon_p$ may lead to $I^* = 0$ since it would be too costly for the society to share the risk burden. In other words, when the cost shock is too high, any attempts at providing indemnity payments may eat up the budget too much and cause a lower value of X overall.

Corollary 0.3 restates Theorem 0.2 in terms of elasticities of ES input supply. It provides a set of sufficient statistics to test whether existing PES programs can achieve higher ES goals by offering indemnity payments. Note that if ES input provision does not introduce risk to the agent, then the condition for both Theorem 0.2 and Corollary 0.3 are not satisfied since the ES input supply would not respond to changes in the indemnity against no-risk.

Applying Theoretical Results

In practice, it is often hard to estimate the indemnity elasticity of supply, a crucial ingredient to evaluate the appeal of offering indemnity, without first having offered indemnity in practice. To address this dilemma, this section aims to offer an approach to estimating the ES supply response to changes in indemnity without needing to observe its outcomes in practice. We first show in (5) that the ES supply response to indemnity ($\frac{\partial x}{\partial I}$) can be expressed as multiple of the ES response to the pay rate ($\frac{\partial x}{\partial p}$).⁷

$$\frac{\partial x}{\partial I} = \frac{\partial x}{\partial p} \left(\mu_\epsilon + \frac{\sigma_{u'}\sigma_\epsilon + x\tilde{\sigma}}{\mu_{u'} [1 - x\bar{A}_\Gamma]} \right) \quad (5)$$

Where $\sigma_{u'}$ and σ_ϵ are the standard deviations of marginal utility and the cost shock respectively, μ_ϵ is expected cost shock, $\mu_{u'}$ is the expected marginal utility, and $\bar{A}_\Gamma = -\frac{\mu_{u''}\Gamma}{\mu_{u'}}$ is a measure of the average absolute risk aversion with a slight difference being that the numerator is the expected value of the second derivative of the utility multiplied by the net marginal return to supplying x , $\Gamma = p - g_x(x, \omega) - \epsilon(1 - I)$. Lastly, $\tilde{\sigma} = Cov(u''\Gamma, \epsilon)$ is the covariance between the cost shock and the product between the second derivative of utility and net marginal return to x . Unfortunately, the product $u''(c)\Gamma$ is a nonmonotonic and discontinuous function meaning that even though all the random variables in the covariance term is derived from the realization of the same random variable, they are still not perfectly correlated. Thus, whether the conditions for Theorem 0.2 to hold in any particular situation is primarily an empirical question.

⁷Note that small x denotes individual ES input while big X denotes aggregate ES input.

In principle, each term in the factor $\frac{\partial x/\partial I}{\partial x/\partial p}$ (implicitly defined in (5)) can be estimated so long as the empirical setting for which it is being applied contains risk associated with the provisioning of ES input x . The reason for this is because without any risk it is impossible to learn the curvature of the utility function since the agent's decision problem is isomorphic to a risk neutral agent who does face risk. Said differently, under a no-risk setting, agents' decisions are determined solely based on payoffs which gives no information on the coefficient of relative risk aversion.

An Estimator for $\overline{A}_\Gamma$

Conditional on the ability to observe relevant moments of ϵ , it is then possible to estimate $\frac{\partial x/\partial I}{\partial x/\partial p}$ simply by estimating our risk aversion parameter $\overline{A}_\Gamma$, I show in Appendix 1 that $\frac{\overline{A}_\Gamma}{\mu_\Gamma}$ acts as a lower bound for $\overline{A} \equiv -\frac{\mu_{u''}}{\mu_{u'}}$ which can be used to estimate the coefficient of absolute risk aversion A which then also provides an estimate for the coefficient of relative risk aversion R . Then one can use the CRRA functional form assumption for utility to finally construct an estimate of the factor from (5).

We follow a similar procedure outlined in Chetty (2006) to derive a formula for $\overline{A}_\Gamma$ expressed as a function of the price elasticity of Hicksian ES input supply and the income elasticity of ES input supply. The intuition behind this approach is outlined in Chetty (2006) which utilized various labor supply elasticities to compute upperbounds for the coefficient of relative risk aversion with the idea being that labor supply responses to wage changes implies the curvature of the utility function which then gives rise the risk aversion parameter.

First, agents' choices on x (assuming an interior optimum) must satisfy the first order condition (2). Taking derivatives of their FOC and rearranging algebraically gives (6).

$$\frac{\partial x}{\partial p} = \frac{\mu_{u'} + \mu_{u''\Gamma} X}{\mu_{u'} g_{xx} - \mu_{u''\Gamma^2}} \quad (6a)$$

$$\frac{\partial x}{\partial w} = \frac{(1 - g_\omega) \mu_{u''\Gamma} - \mu_{u'} g_{x\omega}}{\mu_{u'} g_{xx} - \mu_{u''\Gamma^2}} \quad (6b)$$

Using Slutsky's decomposition (shown in Appendix 2) for compensated ES input supply (x_c)

$$\frac{\partial x_c}{\partial p} = \frac{\partial x}{\partial p} - \frac{\partial x}{\partial \omega} x \quad (7)$$

then the ratio of the substitution effect and income effect is given by (8)

$$\frac{\partial x_c / \partial p}{\partial x / \partial \omega} = \left(\frac{\partial x / \partial p}{\partial x / \partial \omega} - x \right) \quad (8)$$

Then combining (6) with (8) and some algebraic manipulation gives our equation for the desired risk aversion metric (9).

$$\overline{A}_\Gamma = \frac{1 + g_{x\omega} \left(x + \frac{\varepsilon_{xcp} \omega}{\varepsilon_{x\omega} p} \right)}{(g_\omega - 1) \left(\frac{\varepsilon_{xcp} \omega}{\varepsilon_{x\omega} p} \right) + x g_\omega} \quad (9)$$

Thus to estimate the RHS factor in parentheses from (5), one can simply treat the lower bound on $\overline{A}$, which is $\frac{\overline{A}_\Gamma}{\mu_\Gamma}$, as their actual coefficient of absolute risk aversion A . Then the relative risk aversion coefficient is $R = Ac$.

Value Added from Indemnity

These results may leave one wondering how big is the value added from having optimal indemnity alongside the standard linear pricing PES contracts? In both scenarios, the regulator's budget constraint will hold with equality in expectation so that the value added V is given by (10) where $X(I^*, p^*)$ represents the aggregate ES input supply under the optimal contract and $X(0, p(0))$ represents the aggregate ES input supply under the conditionally optimal contract.

$$V = B\left(X(I^*, p^*)\right) - B\left(X(0, p(0))\right) \quad (10)$$

Equation (10) implies that the gain from optimal indemnity PES relative to the conditionally optimal PES is a function of two determinants. First, if the curvature of the expected social benefit function around the neighborhood of $X(0, p(0))$ and $X(I^*, p^*)$ is high, then the value added would be higher. Second, if the difference $X(I^*, p^*) - X(0, p(0))$ is high then so too would be the value added. What makes the difference between these two ES input supplied great has to do with risk tolerance. This boils down to how much can offering indemnity payments allow the regulator to relax $p(I)$, the conditionally optimal pay rate?

To show this, I exploit the fact that was mentioned above which is that the regulator's expenditure under both the optimal and conditionally optimal linear contracts are equal ($p_0 X_0 = X^*(p^* + \mu_\epsilon I)$). This implies that the ratio between the ES input supply under the conditionally optimal contract and that under the optimal contract is given by (11). The lower this ratio is, the higher is the gain from going from conditional optimal contract

to the optimal one. Further, one can see that the difference in ES supplied under the two contracts increases (i.e., the ratio in X 's decreases) as the difference in the optimal prices diverge under the two contracts (i.e., the ratio in p 's decreases). Setting (6a) equal to (6b) and solving for dp/dI gives (12).

$$\frac{X_0}{X^*} = \frac{p^*}{p_0} + \frac{\mu_\epsilon I^*}{p_0} \quad (11)$$

$$\frac{dp}{dI} = -\frac{\frac{\partial x}{\partial I}(p + \mu_\epsilon I) + x\mu_\epsilon}{\frac{\partial x}{\partial p}(p + \mu_\epsilon I) + x} \quad (12)$$

Differences in the optimal prices increase when the response of ES input supply to increases in I are sufficiently greater than the response of ES input supply to decreases in p . In other words, the degree to which the condition in Theorem 0.2 holds. Hence, estimating $\partial x/\partial I$ using estimates for $\partial x/\partial p$ will answer both whether it is optimal to utilize indemnity and how much more value it could bring.

Additionality and Poverty Alleviation

In practice, private agents may produce strictly positive levels of the ES input in the absence of a PES program and therefore there is considerable interest in the literature and policy arena on the idea of additionality. That is, researchers and policy makers want to make sure that the ES inputs being contracted for are *additional* and hence would not have been procured in the absence of a PES program. It is often very costly to monitor additionality for all program applicants which could render simple linear contracts, like the one proposed above, inefficient. There are many papers that try to create a mechanism to generate a

separating equilibrium that is efficient (Mason and Plantinga, 2013). However, this is only necessary when the regulator cannot observe types which is something of the opposite extreme. There are characteristics of households that are cheaply observable such as income or wealth. Then a natural question is when does additionality decrease (or increase) with wealth? Policy makers are often interested in this as this directly answers the question of when is it optimal to have the dual objective of increasing ecosystem services and providing poverty alleviation. Additionality can be defined as

$$\alpha_i = x_i^* - x_i^{bau} \quad (13)$$

It is clear from (13) that additionality decreases with wealth if and only if (14) holds.

$$\frac{\partial x_i^*}{\partial \omega_i} \leq \frac{\partial x_i^{bau}}{\partial \omega_i} \quad (14)$$

Without functional form assumptions, however, evaluating (14) is quite difficult. Instead, Theorem 0.4 relies on the intuition that in order for additionality to decrease in wealth, the slope of the x^{bau} curve, when plotted against ω , cannot be too flat (shown in Figures 1 and 2). A sufficient condition for (14) to hold is for the x^* curve to be decreasing in ω while the x^{bau} curve is increasing. However, neither curve is necessarily required to be increasing or decreasing. What is necessary is for the difference between the two to be decreasing in ω , hence the restriction the slope of the x^{bau} curve.

Theorem 0.4. *Take PES contract (I, p) as given. Additionality decreases in wealth (poorer households have higher additionality) if the change in the business-as-usual level of ES input*

in response to a change in wealth is above some lower bound. Specifically, $\frac{\partial \alpha}{\partial \omega} < 0$ if and only if

$$\frac{\partial x^{bau}}{\partial \omega} > \frac{\mu_{A\Gamma}(h_\omega - 1) - h_{\alpha\omega}}{\mu_{A\Gamma}h_\alpha + \mu_{A\Gamma^2}}$$

where $A = -\frac{u''(c)}{u'(c)}$ is the coefficient of absolute risk aversion, $\Gamma = p - h_\alpha - \epsilon(1 - I)$ is the random marginal return from chosen additionality α , $h(\alpha, \omega)$ is the deterministic cost function in terms of additionality α so that h_α , h_ω , and $h_{\alpha\omega}$ are all the partials with respect to the subscripts. The terms μ simply denote the expectation of the subscripted variables.

See proof on page 26.

Theorem 0.4 states that in order for additionality to decrease in wealth, i.e., poorer agents having higher additionality, the business-as-usual ES input supply must increase in wealth beyond some threshold. In other words, the ES input supplied for poorer agents, in the absence of PES incentives, must be lower than those of wealthier agents and this difference must exceed a threshold. The reason for why a bound is needed on the slope of x^{bau} only is because $\partial x^*/\partial \omega$ has the exact same functional form as $\partial x^{bau}/\partial \omega$ but the two are evaluated at different values for both x and (I, p) . One important take away is that it is not sufficient for the marginal cost of ES input supply to be simply decreasing in wealth in order for additionality to decrease in wealth. That is because the if the marginal cost decreases in wealth, then both the business-as-usual and the PES input supplied will decrease in wealth too leaving the additionality ambiguous.

Lastly, the bound on $\partial x^{bau}/\partial \omega$ in Theorem 0.4 can be estimated if one could estimate aspects of the cost function and relevant elasticities from (9). However, one would still need to be

able to gather sample data on x^{bau} which is often the difficult part. At the very least, the approach outlined in this paper can be used to motivate such an effort to gather this data. Doing so can go a long way to answer the question of when it is optimal for the regulator to pursue the dual objectives of promoting environmental quality and poverty alleviation.

Discussion

This paper shows that a regulator should couple indemnity payments with a standard linear pricing PES whenever the ES supply function is sufficiently more responsive to indemnity than it is to pay rate. Further, the social gain from switching from the conditionally optimal contract to the optimal contract is related one to one with the ratio between the required pay rate under the two contracts. Finally, the theoretical results indicate that targeting low wealth households with a PES contract as a means achieving both poverty alleviation and environmental improvement can be optimal if the slope of the business-as-usual ES supply curve (plotted against wealth) is sufficiently steep.

For practitioners, I show how all of the questions stated above could be answered based on each context by estimating cost functions, a risk aversion parameter $\bar{A}_\Gamma$, and the price response of ES supply $\frac{\partial x}{\partial p}$. The procedure is similar to Chetty (2006) and one important advantage of the approach outlined in this paper is that one can estimate the indemnity response of ES supply without needing to first have indemnity implemented in practice.

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Figures

Figure 1: Additionality Increasing in Wealth

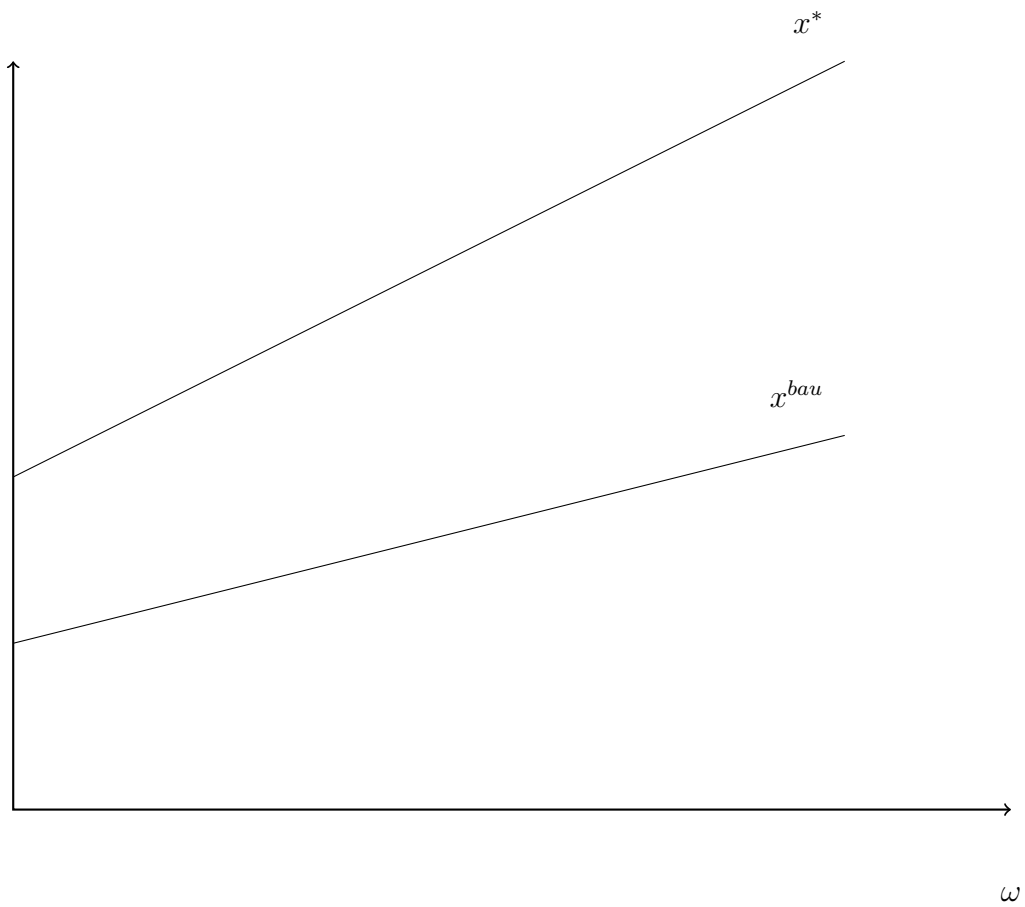
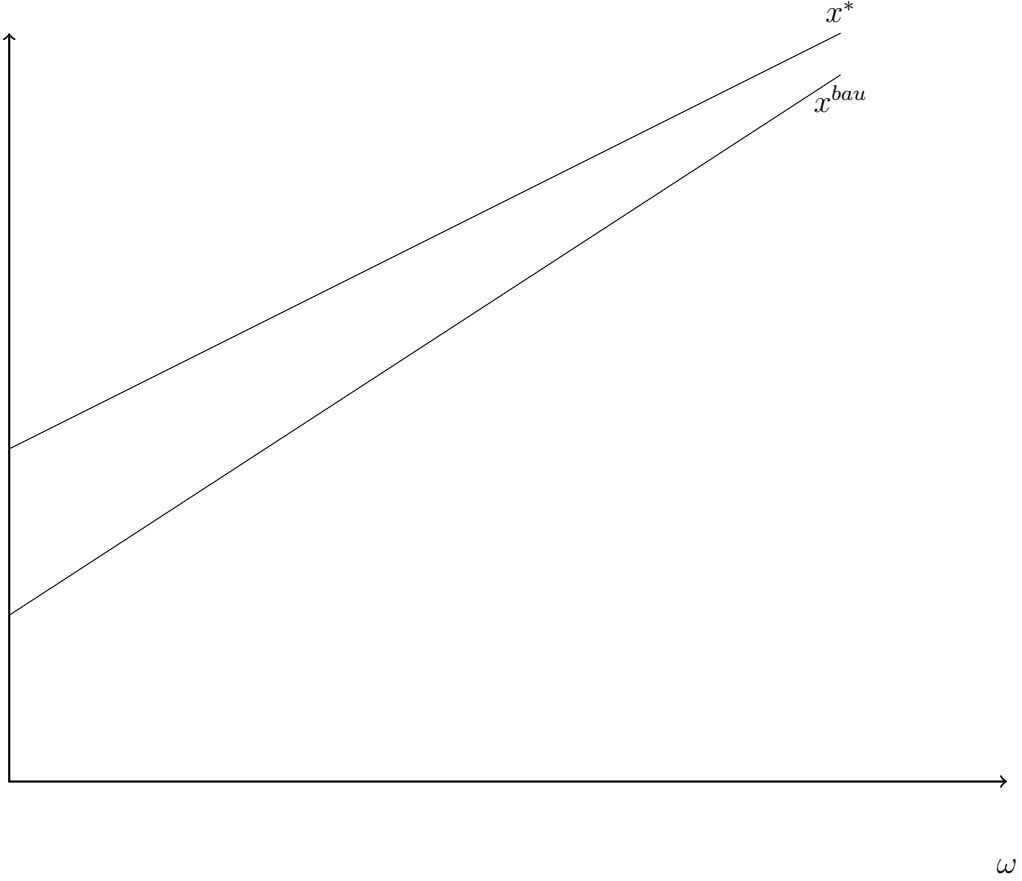


Figure 2: Additionality Decreasing in Wealth



Appendix: Proofs

Proof of Proposition 0.1. To show that x^* is non-decreasing in I , we can invoke the Milgrom-Shannon Monotonicity Theorem by simply showing that $\mathbb{E} \left[u \left(c(x, I) \right) \right]$ is single crossing in (x, I) . Suppose $x' > x$ and $I' > I$. Want to show that (i) $\mathbb{E} \left[u \left(c(x', I) \right) - u \left(c(x, I) \right) \right] \geq 0$ implies $\mathbb{E} \left[u \left(c(x', I') \right) - u \left(c(x, I') \right) \right] \geq 0$ and that (ii) $\mathbb{E} \left[u \left(c(x', I) \right) - u \left(c(x, I) \right) \right] > 0$ implies $\mathbb{E} \left[u \left(c(x', I') \right) - u \left(c(x, I') \right) \right] > 0$.

Let $\mathbb{E} \left[u \left(c(x', I) \right) - u \left(c(x, I) \right) \right] \geq 0$.

$$\iff \mathbb{E} [c(x', I) - c(x, I)] \geq 0 \quad (15)$$

since $u(\cdot)$ is a monotonic transformation of $c(\cdot)$. Rewriting gives

$$\iff \omega + x'p - g(x') - x'\mu_\epsilon(1 - I) - [\omega + xp - g(x) - x\mu_\epsilon(1 - I)] \geq 0$$

$$\iff p(x' - x) - [g(x') - g(x)] - \mu_\epsilon(1 - I)(x' - x) \geq 0$$

Thus we have

$$\implies p(x' - x) - [g(x') - g(x)] - \mu_\epsilon(1 - I')(x' - x) \geq 0$$

$$\iff \mathbb{E} \left[u \left(c(x', I') \right) - u \left(c(x, I') \right) \right] \geq 0$$

The proof for part (ii) is almost identical to that of (i) except with strict inequalities.

To show that x^* is non-decreasing in p follows an almost exact procedure. $\square$

Proof of Theorem 0.2. The goal is to show that the planner can achieve a higher level of indirect social value when $I = 0$ is relaxed. First, let I be a parameter in the planner's problem. The planner then solves the Lagrangian (16) where ψ is the multiplier (distinct

from the multiplier λ in (3)).

$$\mathcal{F} = \max_p B(X) + \psi(G - X(p + \mu_\epsilon I)) \quad (16)$$

The goal can be equivalently stated as wanting to show that $\frac{\partial \mathcal{F}}{\partial I} > 0$. We can use the Envelope Theorem to characterize this partial.

$$\begin{aligned} \frac{\partial \mathcal{F}}{\partial I} &= B'(X) \frac{\partial X}{\partial I} - \psi \left(\frac{\partial X}{\partial I} (p + \mu_\epsilon I) + X \mu_\epsilon \right) \\ &= \frac{\partial X / \partial I}{\partial X / \partial p} \left[B'(X) \frac{\partial X}{\partial p} - \psi \frac{\partial X}{\partial p} (p + \mu_\epsilon I) \right] - \psi X \mu_\epsilon \end{aligned} \quad (17)$$

The first order condition in (16) is given by $B'(X) \frac{\partial X}{\partial p} - \psi \frac{\partial X}{\partial p} (p + \mu_\epsilon I) = \psi X$ so then we can write

$$\frac{\partial \mathcal{F}}{\partial I} = \left(\frac{\partial X / \partial I}{\partial X / \partial p} - \mu_\epsilon \right) \psi X \quad (18)$$

Since every term is positive, then we have that $\frac{\partial \mathcal{F}}{\partial I} > 0$ if and only if $\left(\frac{\partial X / \partial I}{\partial X / \partial p} - \mu_\epsilon \right) > 0$.

Hence, the result. $\square$

Proof of Corollary 0.3. Theorem 0.2 says that $I^* > 0$ if and only if

$$\frac{\partial X}{\partial I} > \frac{\partial X}{\partial p} \mu_\epsilon$$

holds at $I = 0$ (thus p is conditionally optimal). Using the same logic, it must be the case that there exists $I_0 > 0$ such that the strict inequality holds at that point as well. Thus we can validly manipulate this condition to preserve the strict inequality to re-frame in terms

of elasticities which gives

$$\varepsilon_I > \varepsilon_p \mu_\epsilon \frac{I_0}{p}$$

□

Proof of Theorem 0.4. For the comparative statics on additionality α , it is easier to start with a reframing of the model where the agent's choice variable is $\alpha = x - x^b$ instead of x where $x^b = x^{bau}$ for simplicity. Thus consumption is given by

$$c = w + p(\alpha + x^b) - h(\alpha, \omega) - (\alpha + x^b)\epsilon(1 - I)$$

where $h(0, \omega) = g(x^b, \omega)$ and $h(\alpha^*, \omega) = g(x^*, \omega)$. Then the optimal additionality α^* is pinned by

$$\mathbb{E}[u'(c)\Gamma] = 0$$

where $\Gamma \equiv \frac{\partial c}{\partial \alpha}$. Then using the Implicit function theorem, we can differentiate this FOC WRT ω to get

$$\frac{\partial \alpha}{\partial \omega} = \frac{\mu_{A\Gamma}(1 - h_\omega + \frac{\partial x^b}{\partial \omega} h_\alpha) + \mu_{A\Gamma^2} \frac{\partial x^b}{\partial \omega} + h_{\alpha\omega}}{-\mu_{A\Gamma^2} - h_{\alpha\alpha}}$$

and since the denominator is negative, then $\frac{\partial \alpha}{\partial \omega} < 0$ if and only if the numerator is positive.

□

1 Appendix: Lower Bound for $\bar{A}$

Start with the definition of $\bar{A}_\Gamma$

$$\bar{A}_\Gamma = -\frac{\mathbb{E}[u''(c)\Gamma]}{\mathbb{E}[u'(c)]}$$

Using definition of covariance gives

$$\begin{aligned}\bar{A}_\Gamma &= -\frac{Cov(u''(c), \Gamma) + \mu_{u''}\mu_\Gamma}{\mathbb{E}[u'(c)]} \\ &= -\frac{\rho_{u'', \Gamma}\sigma_{u''}\sigma_\epsilon + \mu_{u''}\mu_\Gamma}{\mathbb{E}[u'(c)]} \\ &= \frac{-\sigma_{u''}\sigma_\epsilon - \mu_{u''}\mu_\Gamma}{\mu_{u'}} \quad (u'' \text{ and } \Gamma \text{ are both decreasing in } \epsilon) \\ &= \bar{A}\mu_\Gamma - \frac{\sigma_{u''}\sigma_\epsilon}{\mu_{u'}}\end{aligned}$$

Hence we have

$$\bar{A} > \frac{\bar{A}_\Gamma}{\mu_\Gamma}$$

2 Appendix: Application of Slutsky

Let $e(I, p, u)$ denote the expenditure function defined in (19)

$$\begin{aligned} e = \min_x \quad & g(x, \omega) - xp + x\epsilon(1 - I) \\ \text{s.t.} \quad & u(c) \geq u \end{aligned} \tag{19}$$

The solution to (19) is known as the compensated (Hicksian) ES input supply function x_c .

Applying the Envelope Theorem gives Sheppard's Lemma.

$$\frac{\partial e}{\partial p} = -x_c$$

Making use of the duality at the optimum produces the equality

$$x_c(I, p, u) = x(I, p, e)$$

Hence

$$\frac{\partial x_c}{\partial p} = \frac{\partial x}{\partial p} - \frac{\partial x}{\partial \omega} x$$

Q.E.D.